NOT MEASUREMENT SENSITIVE
MIL-P-85856/38
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS ZIRCONIUM OXIDE, TYPE II, CLASS 2, COMPOSITION H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

SiO_2	CaO	Al_2O_3	TiO_2	Fe_2O_3	MgO	ZrO_2
0.8 max.	4.5 5.5	0.6 max.	0.4 max.	0.3 max.	0.9 max.	91.5 min.

Particle Size/Distribution

+140 Mesh (+106 µm)	1.0 max.
+200 Mesh (+75 µm)	15.0 max.
+325 Mesh (+45 µm)	75.0 min.

Sprayed Coating Properties

Bond Strength	17.3 MPa (2500 psi) min.
Hardness	Rockwell 15N, 72 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 10% max.

Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 12.0 lb/hr
55% min for A/H_2 @ 6.0 lb/hr

AMSC N/A FSC 3439

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Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

Pratt & Whitney Aircraft	PWA 1312
Rolls-Royce Ltd.	MSRR 9507/18
Allison Gas Turbine	EMS 56720C

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/38)

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